

Group assessment

- Group Assignment

- Contribution to overall mark: 30%.

The weekly reading assignments account to 13 marks of your final assignment mark. Students that submit 65% (or more) of the weekly reading assignments are awarded 13 marks in the group coursework; students that submit less than 65% will have the mark adjusted by the proportion of total assignments submitted.

there are 9 reading assignments

READING ASSIGNMENTS ALREADY STARTED

- Enrol in a group (27/02)- 4 to 5 students per group
- Coursework submission week 10
- Coursework discussed on week 5
- Research Consent
- Formulae & Excel videos

$$FV = PV(1 + r)^t$$

Present and Future Value of Stream of Cash Flows and Interest rates

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Future values and compound interest rates

Example-You have £100 in a bank account and banks are currently paying an interest rate of 6% per year on deposits.

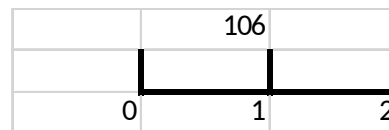
- a. What will the interest be in the end of year 1?

$$100 * 6\% = 100 * 0.06 = 6$$

- b. What will the value of your investment be at the end of year 1?

$$100 + 6 = 106$$

- c. What if you leave the money in the bank for 1 more year?



$$106 * 6\% + 106 = 112.36$$

Future values and compound interest rates

$$FV = PV(1 + r)^t$$

$$PV = 100 \quad t = 2 \quad r = 6\%$$

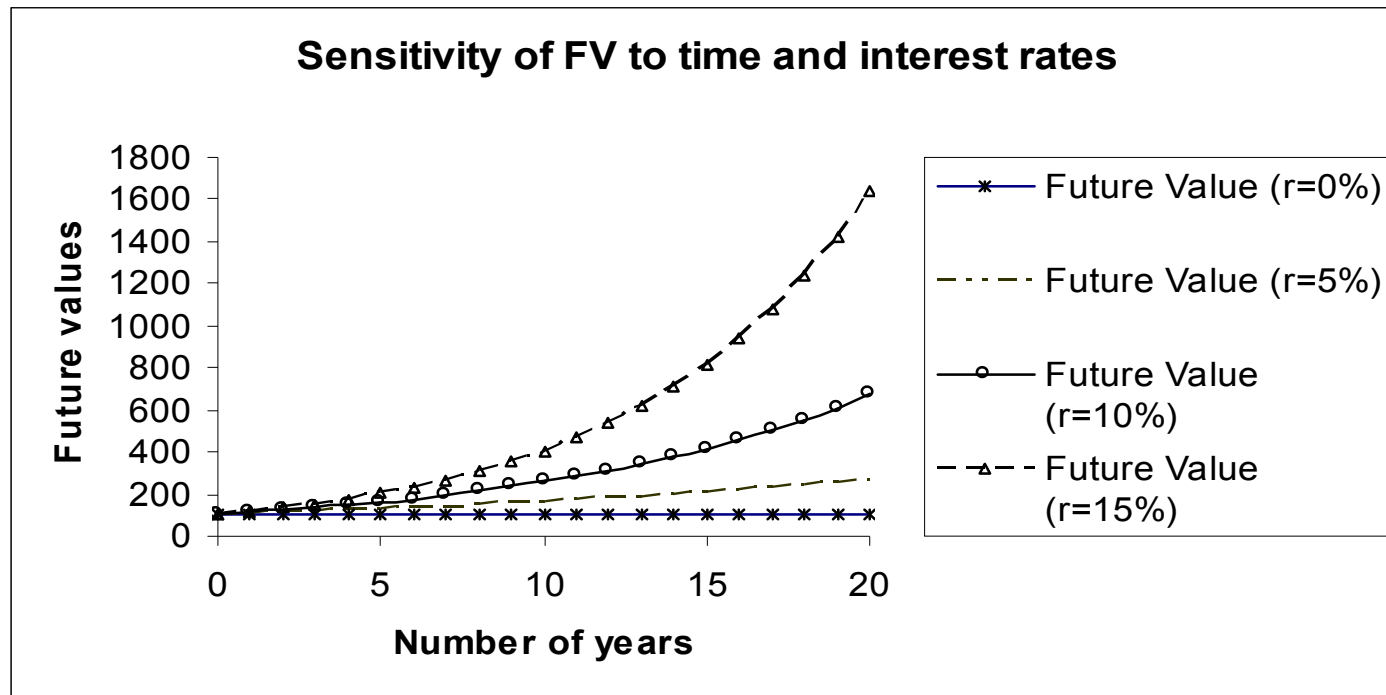
$$FV = 100(1 + 6\%)^2 = 112.36$$

- Compound interest: interest earned on interest (also).
- Simple interest: interest earned only on the original investment

Table A-1 Future Value Interest Factors for One Dollar Compounded at k Percent for n Periods: $FVIF_{k,n} = (1 + k)^n$

Period	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	20%	24%	25%	30%
1	1.0100	1.0200	1.0300	1.0400	1.0500	1.0600	1.0700	1.0800	1.0900	1.1000	1.1100	1.1200	1.1300	1.1400	1.1500	1.1600	1.2000	1.2400	1.2500	1.3000
2	1.0201	1.0404	1.0609	1.0816	1.1025	1.1236	1.1449	1.1664	1.1881	1.2100	1.2321	1.2544	1.2769	1.2996	1.3225	1.3456	1.4400	1.5376	1.5625	1.6900
3	1.0303	1.0612	1.0927	1.1249	1.1576	1.1910	1.2250	1.2597	1.2950	1.3310	1.3676	1.4049	1.4429	1.4815	1.5209	1.5609	1.7280	1.9066	1.9531	2.1970
4	1.0406	1.0824	1.1255	1.1699	1.2155	1.2625	1.3108	1.3605	1.4116	1.4641	1.5181	1.5735	1.6305	1.6890	1.7490	1.8106	2.0736	2.3642	2.4414	2.8561
5	1.0510	1.1041	1.1593	1.2167	1.2763	1.3382	1.4026	1.4693	1.5386	1.6105	1.6851	1.7623	1.8424	1.9254	2.0114	2.1003	2.4883	2.9316	3.0518	3.7129
6	1.0615	1.1262	1.1941	1.2653	1.3401	1.4185	1.5007	1.5869	1.6771	1.7716	1.8704	1.9738	2.0820	2.1950	2.3131	2.4364	2.9860	3.6352	3.8147	4.8268
7	1.0721	1.1487	1.2299	1.3159	1.4071	1.5036	1.6058	1.7138	1.8280	1.9487	2.0762	2.2107	2.3526	2.5023	2.6600	2.8262	3.5832	4.5077	4.7684	6.2749
8	1.0829	1.1717	1.2668	1.3686	1.4775	1.5938	1.7182	1.8509	1.9926	2.1436	2.3045	2.4760	2.6584	2.8526	3.0590	3.2784	4.2998	5.5895	5.9605	8.1573
9	1.0937	1.1951	1.3048	1.4233	1.5513	1.6895	1.8385	1.9990	2.1719	2.3579	2.5580	2.7731	3.0040	3.2519	3.5179	3.8030	5.1598	6.9310	7.4506	10.604
10	1.1046	1.2190	1.3439	1.4802	1.6289	1.7908	1.9672	2.1589	2.3674	2.5937	2.8394	3.1058	3.3946	3.7072	4.0456	4.4114	6.1917	8.5944	9.3132	13.786
11	1.1157	1.2434	1.3842	1.5395	1.7103	1.8983	2.1049	2.3316	2.5804	2.8531	3.1518	3.4785	3.8359	4.2262	4.6524	5.1173	7.4301	10.657	11.642	17.922
12	1.1268	1.2682	1.4258	1.6010	1.7959	2.0122	2.2522	2.5182	2.8127	3.1384	3.4985	3.8960	4.3345	4.8179	5.3503	5.9360	8.9161	13.215	14.552	23.298
13	1.1381	1.2936	1.4685	1.6651	1.8856	2.1329	2.4098	2.7196	3.0658	3.4523	3.8833	4.3635	4.8980	5.4924	6.1528	6.8858	10.699	16.386	18.190	30.288
14	1.1495	1.3195	1.5126	1.7317	1.9799	2.2609	2.5785	2.9372	3.3417	3.7975	4.3104	4.8871	5.5348	6.2613	7.0757	7.9875	12.839	20.319	22.737	39.374
15	1.1610	1.3459	1.5580	1.8009	2.0789	2.3966	2.7590	3.1722	3.6425	4.1772	4.7846	5.4736	6.2543	7.1379	8.1371	9.2655	15.407	25.196	28.422	51.186

Future value sensitivity to interest rate and time



Present values

$$FV = PV(1 + r)^t$$

- **Present value:** value today of a future cash flow.
- The calculation of the present value is the reverse of the calculation of the future value.

Example - How much should you invest now at an interest rate of 6%, to produce £112.36 at the end of 2 years?

100

Present Values

$$FV = PV(1+r)^t \quad FV = 112.36 \quad t = 2 \quad r = 6\% = 0.06 \quad PV = ?$$

$$PV = \frac{FV}{(1+r)^t}$$

$$PV = \frac{112.36}{(1+6\%)^2} = 100$$

$$PV = FV \frac{1}{(1+r)^t}$$

- Present values are always calculated using compound interest rates
- Discount factor: present value of a £1 future payment

Finding the time

$$FV = PV(1 + r)^t$$

A financial manager may need to know for how long must he/she do an investment.

Example - You've deposited today £100 in a bank.
Knowing that the bank pays an interest rate of 4% per year, how long will it take to double your money?

$PV=100$; $FV=200$; $r=4\%$; $t=?$

Finding the time

PV=100; FV=200; r=4%; t=?

$$FV = PV(1 + r)^t$$

$$200 = 100(1 + 4\%)^t \Rightarrow 2 = 1.04^t$$

$$\Rightarrow \log(2) = \log(1.04^t)$$

$$\log(a^b) = b * \log(a)$$

$$\Rightarrow \log 2 = t * \log 1.04 \Rightarrow t = \frac{\log 2}{\log 1.04}$$

$$t = 17.763$$

Finding the interest rate

The financial manager may want to know what is the rate of return that he/she will get from an investment.

Example - Coca-cola issued financial securities, each of which promised to pay £1,000 at the end of 25 years. You bought today one of these securities for £129.20. What is the interest rate?

$FV=1000$; $PV=129.2$; $t=25$; $r=?$

Finding the interest rate

$$FV = PV(1 + r)^t$$

$$FV=1000; PV=129.2; t=25; r=?$$

$$1000 = 129.2(1 + r)^{25} \Rightarrow 7.7399 = (1 + r)^{25}$$

$$\Rightarrow 7.7399^{1 * \frac{1}{25}} = (1 + r)^{\frac{1}{25} * 25}$$

$$\Rightarrow 7.7399^{\frac{1}{25}} = (1 + r)^1$$

$$\Rightarrow r = 0.085299 = 8.5299\%$$

Multiple cash flows

- So far we have only considered a single cash flow. Most real-world investments will involve a stream of cash flows.
- The present value of a stream of cash flows equals the sum of the present value of each of the cash flows.

The Present Value of a Stream of Cash Flows

Suppose that you bought a computer today and you've paid a deposit of £100. The remaining will be paid with 2 instalments, 1 of £250 in 1 year and 1 of £120 in 2 years time. The shop charges an interest of 4% on credit sales. What is the computer's price?

100	250	120
0	1	2

$$PV = \frac{250}{(1 + 4\%)^1}$$

$$PV = \frac{FV}{(1 + r)^t}$$
$$r = 4\%$$

$$PV = \frac{120}{(1 + 4\%)^2}$$

$$PV = 100 + \frac{250}{1.04^1} + \frac{120}{1.04^2} = 451.331$$

Level cash flows: perpetuities and annuities

- Annuity: equally spaced level stream of cash flows, with a finite maturity.
- Perpetuity: Stream of cash flows which last for ever.

If the cash flows are constant we can get to simplified formulas

Or if they grow at a constant rate

Perpetuities

100	C	C	C
0	1	2	

$$Pv(perpetuity) = \frac{C}{r}$$



- The perpetuity formula gives us the value of a regular stream of payments starting 1 period from now.
- All cash are at period end



Important

Perpetuities

100	C	C	C
0	1	2	

Example- Suppose that you invest £100 in a bank and leave this money in the bank forever. Every year you will withdraw the interest at an interest rate of 5% leaving the £100 unchanged, how much interest can you withdraw from the bank every year?

$$C = 100 * 5\% = 5$$

$$PV = 100$$

$$r = 5\%$$

$$C = ?$$

Growing Perpetuity

	C	C(1+g)	C(1+g) ²	
0	1	2	3	...

1	C
2	C(1+g) ¹
3	C(1+g) ²
...	...

$$P_V = \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \dots = \frac{C}{r-g}$$

$$P_{V_{\text{growing perpetuity}}} = \frac{C}{r-g}$$

g is the growth rate (assumed to be constant)

How to value annuities

	C	C	C	C	C	C	C
0	1	2	3	4	5		40

$$Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

- The formula gives us the value of a regular stream of payments starting 1 period from now.
- All cash are at period end

$$\text{Finding } PV \text{ } Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

If you make annual payments of £200 for 3 years to pay for a pc bought today (starting 1 year from now) ; what is the price of the pc if the interest rate you pay on credit is 9% per year?

	200	200	200
0	1	2	3

$$C = 200$$

$$r = 9\%$$

$$n = 3$$

$$PV = ?$$

$$\text{Price} = PV = 200 \left[\frac{1}{9\%} - \frac{1}{9\% * 1.09^3} \right] = £506.258$$

Growing Annuity

If we have an annuity which cash flows grow at a rate g

	C	$C(1+g)$	$C(1+g)^2$
0	1	2	3

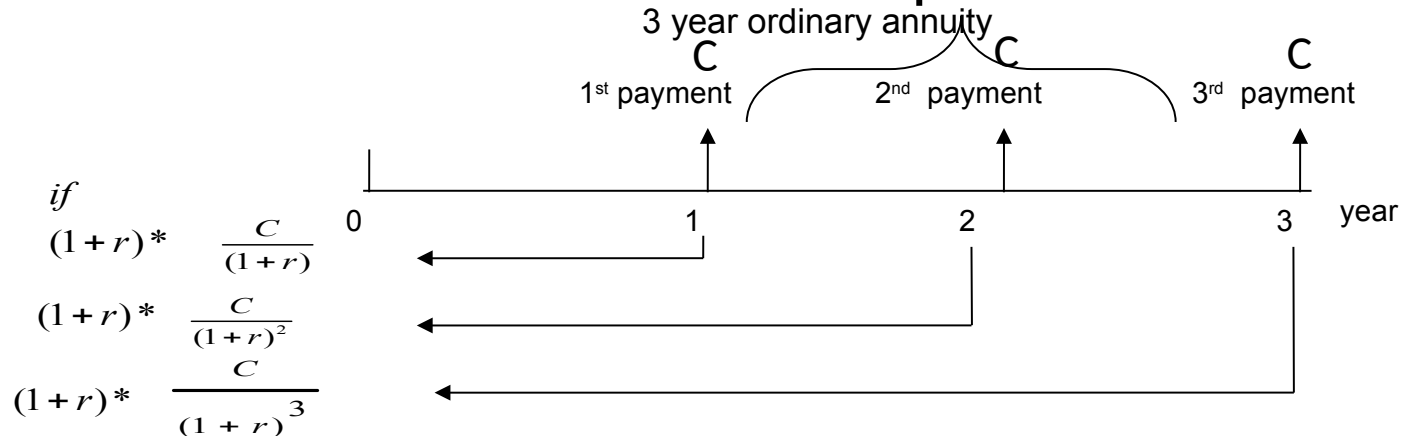
1 C
 2 $C(1+g)^1$
 3 $C(1+g)^2$

$$Pv(\text{annuity growing at rate } g) = C \left[\frac{1 - \left(\frac{1+g}{1+r} \right)^n}{r - g} \right]$$

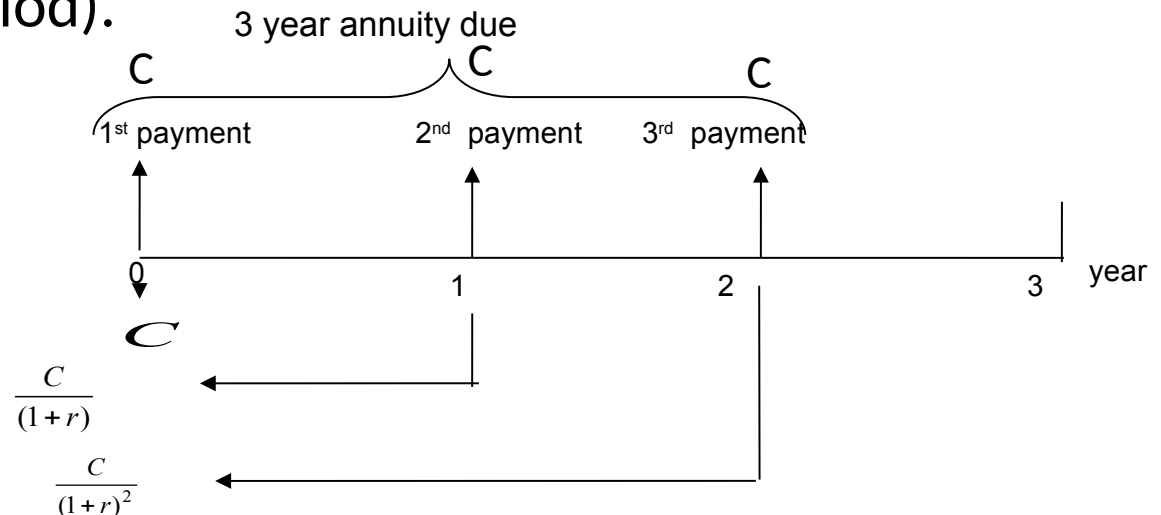
$$PV = \frac{FV}{(1 + r\%)^t}$$

Ordinary Annuity vs Annuity Due

- The perpetuity and annuity formulas assume that the payments are done at the **end** of the period

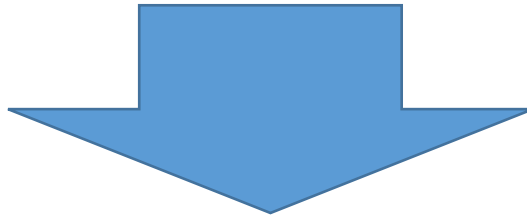


- Annuity due: level stream of cash flows which start immediately (**beginning** of the period).



Relation between the Present values of an ordinary annuity and an annuity due

$$\text{Pv of an annuity due} = \text{Pv of an annuity} * (1 + r)$$



Thus if you are calculating the present value of an annuity due just use the normal formula for the present value of an annuity and then multiply it by $(1+r)$

Finding the present value of an annuity when the stream of cash flows starts later in time

- **Important:** the formula of the present value of an annuity (also true for perpetuities) assumes that the stream of cash flows starts 1 period from now. When this is not the case we still have to discount the value to moment 0.

$$Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

In order to pay for a machine which was just bought, the financial manager of XYZ plc agreed with the seller to pay £2000 for 10 years. The interest rate is 6%, what is the price of the machine knowing that the 1st payment will be done 3 years from now?

C=2000
N=10
r=6%

		2000	2000	2000		2000	2000
1	2	3	4	5		11	12

		2000	2000	2000		2000	2000
1	2	3	4	5		11	12

Finding the present value of an annuity when the stream of cash flows starts later in time

$$Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

C=2000

N=10

$r=6\%$

$$P_v = 2000 \left[\frac{1}{6\%} - \frac{1}{6\%(1 + 6\%)^{10}} \right] = 14720.173$$

14720.173

2000 2000 2000 ... 2000 2000

0 1 2 3 4 5 ... 11 12

$$P_v = \frac{FV}{(1+r)^t} = \frac{14720.173}{1.06^2} = 13100.902$$

Future value of an annuity

$$FV = PV(1 + r)^t$$

The present value of an annuity gives us the value of the stream of cash flows at the beginning of the time period. Conversely, the future value will give us the value at the end.

Imagine that you deposit in a bank £3000 at the end of every year in order to buy a car. If your savings earn interest of 8% a year, how much will you have in the bank at the end of 4 years?

	3000	3000	3000	3000
0	1	2	3	4

$$FV = 3000(1 + 8\%)^3 + 3000(1 + 8\%)^2 + 3000(1 + 8\%) + 3000 = 13518.336$$

How to calculate the future value of an annuity

- Calculate the future value of each cash flow and sum them.
- Calculate the present value of the stream of cash flows (using the Pv of an annuity formula) and then calculate the future value, this results in:

$$Fv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right] (1+r)^n = C \frac{(1+r)^n - 1}{r}$$

Future value of an annuity

Imagine that you deposit in a bank £3000 at the end of every year in order to buy a car. If your savings earn interest of 8% a year, how much will you have in the bank at the end of 4 years?

	3000	3000	3000	3000
0	1	2	3	4

$$C = 3000$$

$$r = 8\%$$

$$n = 4$$

$$FV = ?$$

$$FV = 3000 \frac{(1 + 8\%)^4 - 1}{8\%} = 13518.336$$

Time Value of Money and Mortgages

$$Price = 250000$$

$$r = 0.41667\%$$

$$N = 30 * 12 = 360$$

$$20\% deposit \rightarrow 50000$$

$$Pv = 200000$$

$$C = ?$$

Suppose that a house costs £250,000, and that the buyer puts down 20% of the purchase price, in cash, borrowing the remaining from a mortgage lender.

What is the appropriate monthly mortgage payment knowing that the interest rate is 0.41667% per month? (Assume that the mortgage is paid in 30 years).

$$Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

$$200000 = C \left[\frac{1}{0.41667\%} - \frac{1}{0.41667\%(1+0.41667\%)^{360}} \right]$$

$$C = 1073.64$$

Table of Amortisation of a Loan

$$Pv = 200000$$

$$r = 0.41667\%$$

$$N = 30 * 12 = 360$$

$$C = 1073.64$$

Month	Beginning of month balance	Interest paid - end of the month	Month-end payment	Amortization of loan	End of month balance
1	200000.00	833.33	1073.64	240.31	199759.69
2	199759.69	832.33	1073.64	241.31	199518.38
3	199518.38	831.33	1073.64	242.32	199276.06
4	199276.06	830.32	1073.64	243.33	199032.74
5	199032.74	829.30	1073.64	244.34	198788.40

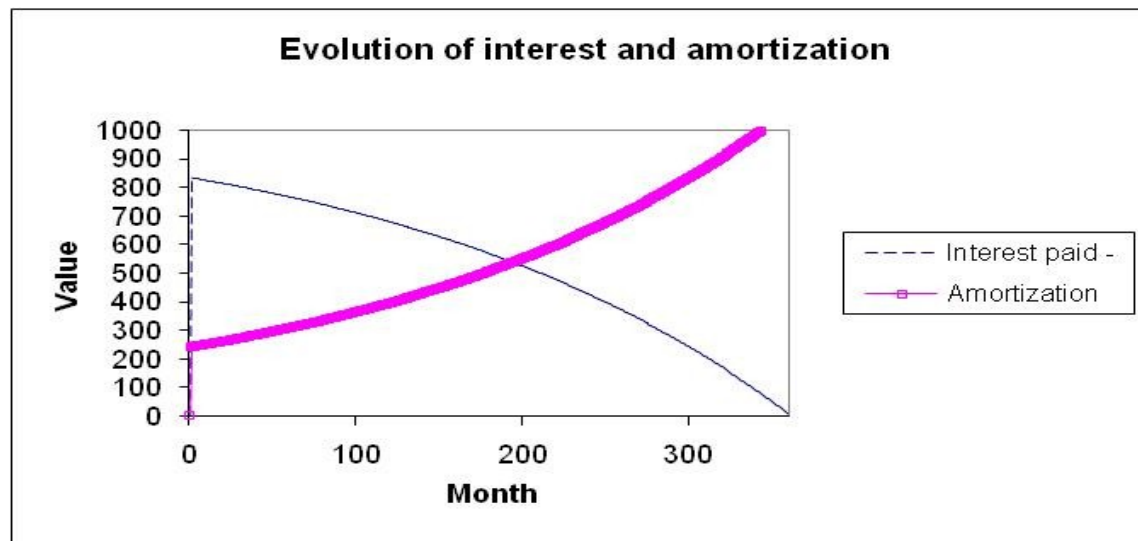
$$200000 * 0.41667\% = 833.33$$

$$1073.64 - 833.33 = 240.31$$

$$200000 - 240.31 = 199759.69$$

Month	Beginning of month balance	Interest paid - end of the month	Month-end payment	Amortization of loan	End of month balance
353	8430.31	35.13	1073.64	1038.52	7391.79
354	7391.79	30.80	1073.64	1042.84	6348.95
355	6348.95	26.45	1073.64	1047.19	5301.76
356	5301.76	22.09	1073.64	1051.55	4250.21
357	4250.21	17.71	1073.64	1055.93	3194.27
358	3194.27	13.31	1073.64	1060.33	2133.94
359	2133.94	8.89	1073.64	1064.75	1069.19
360	1069.19	4.45	1073.64	1069.19	0.00

Mortgage amortisation



The effect of compounding on interest rates

- A: Suppose a bank quotes a 5% yearly rate compounded semi-annually and they tell you that this means you get interest of 2.5% every 6 months.
- B: Is this the same as getting interest of 5% per year?

NO. Say you put £1 in the bank for 1 year:

$$Fv(2.5\%)=? \quad 1 * (1 + 2.5\%)^2 = 1.0506 \quad FV = PV(1 + r)^t$$

$$Fv(5\%)=? \quad 1 * (1 + 5\%) = 1.05$$

Extra examples

Finding C

$$Pv(annuity) = C \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

Bill Gates had his wealth estimated in early 2002 at about \$52 billion. How much could Mr Gates spend yearly in 40 more years of luxury and excitement if he were to devote his entire money to those purposes?

Assume that his money is invested at 9% interest. $PV = 52bn$

52bn	C	C	C	C	C	C	C
0	1	2	3	4	5		40

$$r = 9\%$$

$$n = 40$$

$$C = ?$$

$$52 = C \left[\frac{1}{9\%} - \frac{1}{9\%(1.09)^{40}} \right] \Rightarrow C = 4.834$$

Extra Example

- In order to pay for a loan taken today you have agreed to pay £40,000 in one year time and from then onwards, for 20 years, payments will increase at an annual rate of 6% . The annual interest rate is 10%, what is the value of the loan?

	40	$40(1+g)$	$40(1+g)^2$				
0	1	2	3	4	5		20

$$C = 40,000$$

$$N = 20$$

$$g = 6\%$$

$$r = 10\%$$

$$PV = ?$$

$$PV(\text{annuity growing at rate } g) = C \left[\frac{1 - \left(\frac{1+g}{1+r} \right)^N}{r - g} \right]$$

$$C = 40,000$$

$$N = 20$$

$$g = 6\%$$

$$r = 10\%$$

$$Pv(\text{annuity growing at rate } g) = C \left[\frac{1 - \left(\frac{1+g}{1+r} \right)^N}{r - g} \right] =$$

$$40,000 \left[\frac{1 - \left(\frac{1 + 6\%}{1 + 10\%} \right)^{20}}{10\% - 6\%} \right] = 523,279.7$$